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Display device

Abstract

A display device is provided. The display device includes a substrate, a light-shielding layer, a plurality of first light-emitting elements, and a plurality of second light-emitting elements. The light-shielding layer is disposed on the substrate. The first light-emitting elements are disposed on the substrate. The second light-emitting elements are disposed between the light-shielding layer and the substrate. One of the second light-emitting elements is alternately disposed between any two adjacent first light-emitting elements.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11092861	12/2020	Cheng et al.	N/A	N/A
2006/0061539	12/2005	Song	345/102	G02F 1/133603
2019/0018287	12/2018	Lüchinger	N/A	C09K 11/665
2020/0166794	12/2019	Cheng	N/A	G02F 1/136209
2020/0286967	12/2019	Kuo et al.	N/A	N/A
2020/0341337	12/2019	Sato	N/A	G02F 1/133603

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111221162	12/2019	CN	N/A
112198716	12/2020	CN	N/A
113025112	12/2020	CN	N/A
I670850	12/2018	TW	N/A

OTHER PUBLICATIONS

Chinese language office action dated Feb. 15, 2024, issued in application No. TW 111137788. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of China Patent Application No. 202210237864.X, filed on Mar. 11, 2022, the entirety of which is incorporated by reference herein.

BACKGROUND

Field of the Invention

(2) The present disclosure relates to a display device, and, in particular, to a display device provided with light-emitting elements in a non-display region.

Description of the Related Art

(3) Electronic devices with display panels, such as smartphones, tablet computers, notebook

computers, monitors, and TVs, have become indispensable necessities in modern society. With the vigorous development of such portable electronic products, consumers have higher expectations for the quality, function, and price of such products.

(4) During the evolution of display devices, in order to meet requirements for high resolution and achieve higher productivity, these display devices are constantly being miniaturized, resulting in many unsolved problems in the process of manufacturing light-emitting devices. While existing display devices generally meet the needs of the user, they are not entirely satisfactory in every respect. Therefore, it is still necessary to improve the structure of display devices in order to manufacture a display device that meets product requirements.

BRIEF SUMMARY

(5) A display device is provided according to some embodiments of the present disclosure. The display device includes a substrate, a light-shielding layer, a plurality of first light-emitting elements, and a plurality of second light-emitting elements. The light-shielding layer is disposed on the substrate. The first light-emitting elements are disposed on the substrate. The second light-emitting elements are disposed between the light-shielding layer and the substrate. One of the second light-emitting elements is alternately disposed between any two adjacent first light-emitting elements.

(6) In order to make the features or advantages of the present disclosure more comprehensible, some embodiments are illustrated hereinafter, and detailed descriptions are provided with reference to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a cross-sectional view illustrating a display device according to some embodiments of the present disclosure.

(2) FIG. 1B is a partially enlarged view illustrating the display device of FIG. 1A according to some embodiments of the present disclosure.

(3) FIG. 2 is a cross-sectional view illustrating a display device according to other embodiments of the present disclosure.

(4) FIG. 3 and FIG. 4 are top views illustrating display devices according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(5) Hereinafter is a detailed description of the display device of the embodiments of the present disclosure. It should be understood that the following description provides many different embodiments for implementing various aspects of some embodiments of the present disclosure. The specific elements and arrangements described below are merely to clearly describe some embodiments of the present disclosure. Of course, these are only used as examples rather than limitations of the present disclosure. Furthermore, similar and/or corresponding reference numerals may be used in different embodiments to designate similar and/or corresponding elements, in order to clearly describe the present disclosure. However, the use of these similar and/or corresponding reference numerals is only for the purpose of simply and clearly description of some embodiments of the present disclosure, and does not imply any correlation between the different embodiments and/or structures discussed.

(6) It should be understood that the drawings of the present disclosure are not drawn to scale, and in fact, the dimensions of elements may be arbitrarily enlarged or reduced in order to clearly represent the features of the present disclosure.

(7) In addition, when referring to “a layer is on or over another layer”, it may refer to the case where the layer is in direct contact with another layer. Alternatively, it may also be the case that the

layer is not in direct contact with another layer. In this case, one or more intermediate layers are disposed between the layer and another layer.

(8) In addition, it should be understood that ordinal numbers such as “first”, “second”, and the like used in the specification and claims are used to modify elements and are not intended to imply and represent the element(s) have any previous ordinal numbers, and do not represent the order of a certain element and another element, or the order of the manufacturing method. The use of these ordinal numbers is only used to clearly distinguish an element with a certain name and another element with the same name. The claims and the specification may not use the same terms, for example, a first element in the specification may be a second element in the claims.

(9) The term “about” used herein generally means within 10%, within 5%, within 3%, within 2%, within 1%, or within 0.5% of a given value or a given range. The value given herein is an approximate value, that is, the meaning of “about” may still be implied without the specific description of “about”. Furthermore, the phrase “a range is greater than or equal to a first value, and the range is less than or equal to a second value” means that the range includes the first value, the second value, and other values in between.

(10) Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skills in the art. It should be understood that these terms, such as those defined in commonly used dictionaries, should be interpreted as having meanings consistent with the relevant art and the background or context of the present disclosure, and should not be interpreted in an idealized or overly formal manner, unless otherwise defined in the embodiments of the present disclosure.

(11) According to embodiments of the present disclosure, an optical microscope (OM), a scanning electron microscope (SEM), a film thickness profiler (α -step), an ellipsometer, or other suitable methods may be used to measure the spacing or distance between elements, or the width, thickness, height, or area of each element. In detail, according to some embodiments, a scanning electron microscope may be used to obtain a cross-sectional structure image including the element to be measured, so as to measure the spacing or distance between elements, or the width, thickness, height, or area of each element.

(12) It should be understood that in the following embodiments without departing from the spirit of the present disclosure, features in several different embodiments may be replaced, combined, and recombined to become other embodiments. As long as the features of the embodiments do not violate the spirit or conflict with each other, they may be arbitrarily recombined and used.

(13) According to some embodiments of the present disclosure, a display device is provided. The display device includes second light-emitting elements disposed between a substrate and a light-shielding layer (i.e., within a non-display region of the display device). The second light-emitting elements are alternately disposed between two adjacent first light-emitting elements in a display region. As a result, the brightness difference between the periphery and the center of the display region of the display device may be reduced, so as to improve the problem of non-uniform brightness of the display device.

(14) It should be understood that besides the display device, the embodiments of the present disclosure may be applied to various electronic devices, such as light-emitting devices, touch-control devices, sensing devices, antenna devices, splicing devices, or a combination thereof, but the present disclosure is not limited thereto. The electronic device may be a bendable or flexible electronic device. The electronic device may include, for example, light-emitting diodes, liquid-crystal, fluorescence, phosphor, other suitable display media, or a combination thereof, but the present disclosure is not limited thereto. The light-emitting diode may include, for example, organic light-emitting diodes (OLEDs), inorganic light-emitting diodes (LED), mini-light-emitting diodes (mini LED), micro-light-emitting diodes (micro-LEDs), quantum dots (QDs) light-emitting diode (such as QLEDs, QDLEDs), other suitable materials, or any arrangement or combination thereof, but the present disclosure is not limited thereto. The display device may include, for

example, a splicing device, but the present disclosure is not limited thereto. The concepts or principles of the present disclosure may also be applied to a non-self-luminous liquid-crystal display (LCD), but the present disclosure is not limited thereto.

(15) The antenna device may be, for example, a liquid-crystal antenna or another type of antenna, but the present disclosure is not limited thereto. The antenna device may include, for example, a splicing-antenna device, but the present disclosure is not limited thereto. It should be noted that the electronic device may be any arrangement or combination thereof above, but the present disclosure is not limited thereto. In addition, the shape of the electronic device may be rectangular, circular, polygonal, a shape with curved edges, or other suitable shapes. The electronic device may have peripheral systems such as a driving system, a control system, a light source system, a shelf system, and the like to support the display devices, the antenna devices, or the splicing devices. The electronic device of the present disclosure may be, for example, a display device, but the present disclosure is not limited thereto.

(16) FIG. 1A is a cross-sectional view illustrating a display device **10** according to some embodiments of the present disclosure. Referring to FIG. 1A, the display device **10** includes a substrate **102** and a light-shielding layer **122** disposed on the substrate **102**. In some embodiments, the light-shielding layer **122** may be disposed on at least one side of the substrate **102**. For example, the light-shielding layer **122** may be disposed on one side, two sides, or three sides of the substrate **102**. In an embodiment, as shown in FIG. 1A, the light-shielding layer **122** may be disposed around the substrate **102**. It should be noted that in FIG. 1A, the portion of the substrate **102** not covered by the light-shielding layer **122** is referred to herein as the “display region R1” of the display device **10**, and the portion of the substrate **102** covered by the light-shielding layer **122** is referred to herein as the “non-display region R2” of the display device **10**.

(17) According to some embodiments, the substrate **102** may include a flexible substrate, a rigid substrate, or a combination thereof, but the present disclosure is not limited thereto. According to some embodiments, the material of the substrate **102** may include glass, quartz, sapphire, ceramic, polyimide (PI), polycarbonate (PC), polyethylene terephthalate (PET), polypropylene (PP), other suitable materials, or a combination thereof, but the present disclosure is not limited thereto. Furthermore, according to some embodiments, the substrate **102** may include a metal-glass fiber composite sheet or a metal-ceramic composite sheet, but the present disclosure is not limited thereto. In addition, the light transmittance of the substrate **102** is not limited, that is, the substrate **102** may be a transparent substrate, a semi-transparent substrate, or a non-transparent substrate. Furthermore, the substrate **102** may be a driving substrate of the display device **10**.

(18) In various embodiments of the present disclosure, the light-shielding layer **122** is defined as a light-shielding material that may prevent the user from observing the light penetration with the naked eye, such as a material with an optical density greater than about 3.5, or a material with light transmittance of approximately 0% for 50 W, 101 m of light according to the standard test method of JIS K 7150. In some embodiments, the material of the light-shielding layer **122** may include black photoresist, black printing ink, black resin, any other suitable light-shielding material, or a combination thereof, but the present disclosure is not limited thereto.

(19) As shown in FIG. 1A, the display device **10** further includes a plurality of first light-emitting elements **104A** and a plurality of second light-emitting elements **104B**. The first light-emitting elements **104A** are disposed on the substrate **102**, and the second light-emitting elements **104B** are disposed between the substrate **102** and the light-shielding layer **122**. In detail, the first light-emitting elements **104A** are disposed in the region of the substrate **102** not covered by the light-shielding layer **122**. That is, the first light-emitting elements **104A** are disposed in the display region R1 of the display device **10**. The second light-emitting elements **104B** are disposed in the region of the substrate **102** covered by the light-shielding layer **122**. That is, the second light-emitting elements **104B** are disposed in the non-display region R2 of the display device **10**.

(20) In some embodiments, the first light-emitting elements **104A** and the second light-emitting

elements **104B** may be mini light-emitting diodes (mini LED), micro light-emitting diodes (micro LED), and light-emitting diodes. Furthermore, in some embodiments, the first light-emitting elements **104A** and the second light-emitting elements **104B** may each emit blue light, red light, green light, white light, or ultraviolet light. In an embodiment, the first light-emitting elements **104A** and the second light-emitting elements **104B** may emit blue light.

(21) In some embodiments, the display device **10** may further include an adhesive layer **106**. The adhesive layer **106** is formed on the substrate **102**, the first light-emitting elements **104A**, and the second light-emitting elements **104B**. The first light-emitting elements **104A** and the second light-emitting elements **104B** are encapsulated in the adhesive layer **106** to prevent moisture or oxygen from damaging the first light-emitting elements **104A** and the second light-emitting elements **104B**.

(22) In some embodiments, the adhesive layer **106** may be transparent or semi-transparent to avoid reducing the light extraction rate of the first light-emitting elements **104A** and the second light-emitting elements **104B**. The adhesive layer **106** may include organic material, inorganic material, or a combination thereof. In some embodiments, the inorganic material may include silicon nitride, silicon oxide, silicon oxynitride, aluminum oxide, any other suitable packaging material, or a combination thereof, but the present disclosure is not limited thereto. In some embodiments, the organic material includes, epoxy resin, acrylic resin such as polymethylmetacrylate (PMMA), benzocyclobutene (BCB), polyimide, polyester, polydimethylsiloxane (PDMS), perfluoroalkoxy (PFA), any other suitable protective material, or a combination thereof, but the present disclosure is not limited thereto. In some embodiments, the adhesive layer **106** may be formed by a suitable method such as chemical vapor deposition (CVD), spin-on coating, printing, or a combination thereof.

(23) In some embodiments, the display device **10** may further include a brightness enhancement film **108**. The brightness enhancement film **108** is disposed on the adhesive layer **106**. The brightness enhancement film **108** enables the light emitted by the first light-emitting elements **104A** and the second light-emitting elements **104B** to form a relatively concentrated light source after passing through the brightness enhancement film **108**. In an embodiment, the brightness enhancement film **108** may be a blue brightness enhancement film.

(24) In some embodiments, the display device **10** may further include a color conversion layer **110** formed on the brightness enhancement film **108**. According to some embodiments, the color conversion layer **110** may include polymer material with color conversion material distributed therein. The color conversion material is such as any suitable quantum dots (QDs) material, phosphor material, or phosphor material. In an embodiment, the color conversion layer **110** includes phosphorescent material. In some embodiments, the polymer material above may be a transparent polymer material. Specifically, the transparent polymer material has a high transmittance for light with wavelengths between 200 nm and 1100 nm. For example, the transparent polymer material may have a light transmittance of greater than 90% or may have a light transmittance of greater than 95%. For example, the transparent polymer material may include polymethyl methacrylate (PMMA), polyethylene terephthalate (PET), polystyrene (PS), polyethylene (PP), nylon (polyamide, PA), polycarbonate (PC), polyimide (PI), polydimethylsiloxane (PDMS), epoxy resin, silicone, and the like, or a combination thereof, but the present disclosure is not limited thereto.

(25) According to some embodiments, when the first light-emitting elements **104A** and the second light-emitting elements **104B** emit blue light or ultraviolet light, suitable quantum dots material, phosphorescent material, or phosphor material may be added to the color conversion layer **110** according to product requirements to emit light with the desired color, such as red, green, or white.

(26) In some embodiments, the display device **10** may further include an optical layer **112** disposed on the first light-emitting elements **104A** and the second light-emitting elements **104B**.

Furthermore, according to some embodiments, the optical layer **112** may be further disposed on the color conversion layer **110**. According to some embodiments, as shown in FIG. 1A, the optical

layer **112** may cover the color conversion layer **110**. In some embodiments, the optical layer **112** may be a diffusion plate to diffuse light passing through the underlying layer to generate a uniform surface light source. In some embodiments, the optical layer **112** may have a single-layer structure or a multi-layer structure. In an embodiment, when the optical layer **112** has a multi-layer structure, the multi-layer structure may be formed by stacking the same single-layer structures. In some embodiments, the brightness enhancement film **108** may be further disposed on the optical layer **112**.

(27) In some embodiments, the display device **10** may further include a light conversion layer **114**, and the light conversion layer **114** is disposed to partially overlap with the optical layer **112**.

According to some embodiments, the light conversion layer **114** partially overlaps with the optical layer **112** in the normal direction of the substrate **102**. In detail, in some embodiments, as shown in FIG. **1A**, the light conversion layer **114** may be disposed on the lower surface **112L** of the optical layer **112**. In an embodiment, the light conversion layer **114** is disposed on a portion of the lower surface **112L** of the optical layer **112**. More specifically, the light conversion layer **114** is disposed on the edge of the lower surface **112L** of the optical layer **112** such that the light conversion layer **114** is disposed to partially overlap with the optical layer **112**. Furthermore, in some embodiments, the light conversion layer **114** may be disposed between the second light-emitting elements **104B** and the optical layer **112**. In some embodiments, the light conversion layer **114** may be further disposed between some of the first light-emitting elements **104A** and the optical layer **112**.

(28) According to some embodiments, the light conversion layer **114** may include light conversion ink. In detail, the light conversion ink may include light conversion material and curing material. For example, in some embodiments, the light conversion material may include any suitable quantum dots material or phosphor material, and the curing material may include (meth)acrylic-based material or epoxy resin-based material. According to some embodiments of the present disclosure, the optical layer **112** may cover the color conversion layer **110**, the first light-emitting elements **104A**, and the second light-emitting elements **104B**. The light emitted by the second light-emitting elements **104B** located in the non-display region **R2** of the display device **10** may pass through the space other than the color conversion layer **110** without passing through the color conversion layer **110**, resulting in a part of the light not being converted into light with the desired color. Therefore, disposing the light conversion layer **114** on the edge of the lower surface **112L** of the optical layer **112** may improve the light conversion efficiency of the second light-emitting elements **104B** located in the non-display region **R2** of the display device **10**, so as to prevent the peripheral of the display region **R1** of the display device **10** from displaying unwanted colors. In some embodiments, the light conversion layer **114** may be coated on a predetermined area (e.g., the edge of the lower surface **112L**) on the lower surface **112L** of the optical layer **112** by inkjet printing.

(29) In FIG. **1A**, the structure formed by the substrate **102**, the first light-emitting elements **104A**, the second light-emitting elements **104B**, the adhesive layer **106**, the brightness enhancement film **108**, the color conversion layer **110**, the optical layer **112**, and the light conversion layer **114** may be collectively referred to herein as the “backlight module **116**”.

(30) Still referring to FIG. **1A**, according to some embodiments, the display device **10** may further include a case **118**, and the backlight module **116** is disposed on the case **118**. In some embodiments, the material of the case **118** may include metal, plastic, ceramic, other suitable materials, or a combination thereof, but the present disclosure is not limited thereto. In some embodiments, the display device **10** may further include a frame member **120**. The frame member **120** is disposed on the case **118** and surrounds the backlight module **116**. According to some embodiments, the material of the frame member **120** may include metal, plastic, or a combination thereof. For example, the metal may include aluminum, stainless steel, or galvanized steel, but the present disclosure is not limited thereto. Furthermore, the plastic may include polycarbonate (PC), acrylonitrile-butadiene-styrene (ABS) copolymer, and the like, or a combination thereof, but the

present disclosure is not limited thereto.

(31) According to some embodiments, as shown in FIG. 1A, the portion of the optical layer **112** not covering the color conversion layer **110** and the portion of the light conversion layer **114** not covering the color conversion layer **110** may be disposed on the frame member **120** and be in contact with the frame member **120** to ensure the light emitted from the second light-emitting elements **104B** may pass through the light conversion layer **114** to avoid the problem of incomplete light conversion.

(32) In some embodiments, the light-shielding layer **122** of the display device **10** may be disposed on the backlight module **116** and the frame member **120**. As described above, the region of the substrate **102** covered by the light-shielding layer **122** may be referred to as the non-display region R2 of the display device **10**, and the second light-emitting elements **104B** are disposed in the non-display region R2. The region of the substrate **102** not covered by the light-shielding layer **122** may be referred to as the non-display region R1 of the display device **10**, and the first light-emitting elements **104A** are disposed in the display region R1.

(33) In some embodiments, the display device **10** may further include a polarizing plate **124**, a polarizing plate **128**, and a liquid-crystal layer **126**. The polarizing plate **124**, the polarizing plate **128**, and the liquid-crystal layer **126** are disposed on the backlight module **116** and the light-shielding layer **122**. As shown in FIG. 1A, the liquid-crystal layer **126** is sandwiched between the polarizing plate **124** and the polarizing plate **128**. In other words, the polarizing plate **124** and the polarizing plate **128** are disposed on opposite sides (i.e., the upper surface and the lower surface) of the liquid-crystal layer **126**.

(34) According to some embodiments, the material of the polarizing plate **124** and the polarizing plate **128** may include polyvinyl alcohol (PVA) or other suitable materials, but the present disclosure is not limited thereto. In other embodiments, the polarizing plate **124** and the polarizing plate **128** may each be a wire grid polarizer (WGP).

(35) According to some embodiments, the liquid-crystal layer **126** may include liquid-crystal molecules (not shown). In some embodiments, the image displayed may be adjusted by applying different electric fields to the liquid-crystal layer **126** to change the alignment direction of the liquid-crystal molecules. In some embodiments, the material of the liquid-crystal layer **126** may include nematic liquid-crystal, smectic liquid-crystal, cholesteric liquid-crystal, blue-phase liquid-crystal, other suitable liquid-crystal materials, or a combination thereof, but the present disclosure is not limited thereto. However, according to other embodiments, the liquid-crystal layer **126** may be replaced with modulation material having different properties (e.g., permittivity) through application of an electric field or other adjustments.

(36) Next, refers to FIG. 1B, which is a partially enlarged view illustrating the display device of FIG. 1A according to some embodiments of the present disclosure. According to some embodiments, the thickness T1 of the light conversion layer **114** may be between about 1 μm and about 10 μm . If the thickness T1 of the light conversion layer **114** is too large (e.g., greater than about 10 μm), the light conversion layer **114** may affect the color of the light passing through the light conversion layer **114** because the material of the light conversion layer **114** may absorb light having a specific wavelength. If the thickness T1 of the light conversion layer **114** is too small (e.g., less than about 1 μm), the light conversion efficiency of the light conversion layer **114** may be too low to achieve the desired light conversion effect.

(37) According to some embodiments, the distance D1 between the outer edge of the light conversion layer **114** and the sidewall of at least one of the substrate **102**, the adhesive layer **106**, the brightness enhancement film **108**, or the color conversion layer **110** is between about 0.3 mm and about 0.5 mm. In other words, the length of the portion of the light conversion layer **114** extending outward from at least one of the substrate **102**, the adhesive layer **106**, the brightness enhancement film **108**, or the color conversion layer **110** is the distance D1 shown in FIG. 1B.

(38) According to some embodiments, the distance D2 between the inner edge of the light-

shielding layer **122** in the normal direction of the substrate **102** and the first light-emitting element **104A** at the outermost periphery of the display region **R1** is about 0.3 mm to about 0.5 mm. If the distance **D2** is too large, for example, greater than about 0.5 mm, the dark band around the display region **R1** due to insufficient brightness is more obvious. If the distance **D2** is too small, for example, less than about 0.3 mm, the disposition space of the second light-emitting element **104B** in the non-display region **R2** is limited. The portion of the substrate **102** covered by the light-shielding layer **122** belongs to the non-display region **R2** of the display device **10**. Therefore, the distance between the boundary of the non-display region **R2** (i.e., the boundary between the display region **R1** and the non-display region **R2**) and the first light-emitting element **104A** at the outermost periphery of the display region **R1** is also the distance **D2**.

(39) On the other hand, according to some embodiments, the distance **D3** between the boundary of the non-display region **R2** (i.e., the inner edge of the light-shielding layer **122**, the boundary of the display region **R1** and the non-display region **R2**) and the second light-emitting element **104B** in the normal direction of the substrate **102** may be between about 0 mm and about 0.2 mm. If the distance **D3** is too large, for example, greater than about 0.2 mm, the dark band around the display region **R1** due to insufficient brightness is more obvious. In an embodiment, when the distance **D3** between the boundary of the non-display region **R2** and the second light-emitting element **104B** is about 0 mm, the sidewall of the second light-emitting element **104B** may be aligned with the boundary of the non-display region **R2**, thereby enhancing the brightness of the periphery of the display region **R1**. That is, the problem of the obvious dark band around the display region **R1** may be alleviated.

(40) As shown in FIG. **1B**, the distance **D4** between the first light-emitting element **104A** and the second light-emitting element **104B** is the sum of the above distance **D2** and distance **D3**. In some embodiments, the distance **D4** between the first light-emitting element **104A** and the second light-emitting element **104B** may be between about 0.3 mm and about 0.7 mm. If the distance **D4** between the first light-emitting element **104A** and the second light-emitting element **104B** is too large, e.g., greater than about 0.7 mm, the dark band around the display region **R1** due to insufficient brightness is more obvious. If the distance **D4** between the first light-emitting element **104A** and the second light-emitting element **104B** is too small, for example, less than about 0.3 mm, sufficient space may not be reserved between the first light-emitting element **104A** and the second light-emitting element **104B** to ensure that the first light-emitting elements **104A** are disposed in the display region **R1** and the second light-emitting elements **104B** are disposed in the non-display region **R2**.

(41) According to some embodiments, the length **L1** from the outer edge to the inner edge of the light conversion layer **114** may be between about 2.15 mm and about 2.55 mm. In some embodiments, as shown in FIG. **1B**, the outer edge of the light conversion layer **114** may be aligned with the edge of the optical layer **112** in the normal direction of the substrate **102**, but the present disclosure is not limited thereto. If the length **L1** from the outer edge to the inner edge of the light conversion layer **114** is too large (for example, greater than about 2.55 mm), the light conversion layer **114** will significantly affect the light emitted by the first light-emitting elements **104A** in the display region **R1** and cause light with undesirable color. Conversely, if the length **L1** from the outer edge to the inner edge of the light conversion layer **114** is too small (for example, less than about 2.15 mm), the light emitted from the second light-emitting elements **104B** without passing through the color conversion layer **110** may not be effectively converted, and resulting in the phenomenon of non-uniform color around the display region **R1** of the device **10**.

(42) Next, referring to FIG. **2**, which is a cross-sectional view illustrating the display device **20** according to other embodiments of the present disclosure. The display device **20** shown in FIG. **2** is similar to the display device **10** shown in FIG. **1**, but an additional light conversion layer **114** is further disposed on the upper surface **112U** of the optical layer **112** in the display device **20**. The material and size of the light conversion layer **114** disposed on the upper surface **112U** of the

optical layer **112** may be the same as those of the light conversion layer **114** disposed on the lower surface **112L** of the optical layer **112**, so the description is omitted. Furthermore, the position of the light conversion layer **114** disposed on the upper surface **112U** of the optical layer **112** may be the same as the position of the light conversion layer **114** disposed on the lower surface **112L** of the optical layer **112**. In other words, the light conversion layer **114** on the upper surface **112U** may be aligned with the light conversion layer **114** on the lower surface **112L** in the normal direction of the substrate **102**.

(43) FIG. **3** and FIG. **4** are top views respectively illustrating the display device **10** and display device **30** according to various embodiments of the present disclosure. It should be noted that the cross-sectional view of FIG. **1A** is taken along the line CC' in FIG. **3**. Furthermore, for the sake of clarity, many elements and layers are omitted in FIG. **3** and FIG. **4** to facilitate the description of the illustrated embodiment.

(44) As shown in FIG. **3**, the second light-emitting elements **104B** are disposed alternately between two adjacent light-emitting elements **104A**. Specifically, the second light-emitting elements **104B** are disposed alternately between two adjacent first light-emitting elements **104A** in the outermost periphery of the display region **R1**. In detail, according to some embodiments, there is an extension line **E1** between any two adjacent first light-emitting elements **104A** in the outermost periphery of the display region **R1** in the A direction, and one of the second light-emitting elements **104B** in the non-display region **R2** has an extension line **E2** in the B direction. In some embodiments, the extension direction of the extension line **E1** and the extension direction of the extension line **E2** are different, so that the extension line **E1** and the extension line **E2** intersect each other. In some embodiments, the extension line **E1** and the extension line **E2** may be perpendicular to each other. When the extension line **E1** and the extension line **E2** are perpendicular to each other, the A direction and the B direction are also perpendicular to each other.

(45) According to some embodiments, as shown in FIG. **3**, the first light-emitting elements **104A** and the second light-emitting elements **104B** may be alternately disposed in the A direction. Likewise, according to some embodiments, the first light-emitting elements **104A** and the second light-emitting elements **104B** may be alternately disposed in the B direction.

(46) According to the embodiments provided by the present disclosure, the second light-emitting elements **104B** in the non-display region **R2** of the display device **10** and the first light-emitting elements **104A** in the display region **R1** of the display device **10** are alternately disposed, which may further alleviate the problem of the non-uniform brightness of the periphery of the display region **R1** and thus avoid the generation of the dark band with insufficient brightness around the display region **R1**.

(47) Still referring to FIG. **3**, according to some embodiments, the second light-emitting elements **104B** are disposed around the first light-emitting elements **104A**. In other embodiments, the second light-emitting elements **104B** may be disposed on at least one side of the first light-emitting element **104A**. When the dark band with insufficient brightness occurs on one side of the periphery of the display region **R1**, the second light-emitting elements **104B** may be disposed on the side to compensate the problem of insufficient brightness. Similarly, if there are dark bands with insufficient brightness around the display region **R1** on multiple sides, the second light-emitting elements **104B** may be disposed on two sides, three sides, or each side.

(48) Furthermore, according to some embodiments, the pitch **P1** between two adjacent first light-emitting elements **104A** and the pitch **P3** between two adjacent second light-emitting elements **104B** may be the same in the B direction. In other embodiments, the pitch **P1** and the pitch **P3** may be different. In the embodiment in which the pitch **P1** and the pitch **P3** are the same, the pitch **P2** between the first light-emitting element **104A** and the second light-emitting element **104B** in the B direction is $\frac{1}{2}$ of the pitch **P1** and the pitch **P3**. Likewise, according to some embodiments, the pitch **P4** between two adjacent first light-emitting elements **104A** and the pitch **P6** between two adjacent second light-emitting elements **104B** may be the same in the A direction. In other embodiments, the

pitch P4 and the pitch P6 may be different. In the embodiment in which the pitch P4 and the pitch P6 are the same, the pitch P5 between the first light-emitting element 104A and the second light-emitting element 104B in the A direction is $\frac{1}{2}$ of the pitch P4 and the pitch P6.

(49) Next, referring to FIG. 4, the display device 30 shown in FIG. 4 is similar to the display device 10 shown in FIG. 3, but the first light-emitting elements 104A in the display region R1 of the display device 30 are alternately disposed. In detail, according to some embodiments, the first light-emitting elements 104A and the adjacent first light-emitting elements 104A are alternately disposed in the A direction, and the first light-emitting elements 104A are also alternately disposed in the B direction with the adjacent first light-emitting elements 104A. Therefore, as shown in FIG. 4, the first light-emitting elements 104A located in the same column or row and the first light-emitting elements 104A adjacent to one column or adjacent to one row are staggered to each other.

(50) It should be noted that although FIG. 3 and FIG. 4 respectively illustrate that the first Light-emitting elements 104A may be disposed on the substrate 102 in different arrangements in the display region R1, the present disclosure is not limited thereto. In other embodiments, the first light-emitting elements 104A may be disposed on the substrate 102 in any manner in the display region R1.

(51) As mentioned above, according to some embodiments of the present disclosure, a display device includes first light-emitting elements located in a display region and second light-emitting elements located in a non-display region. The second light-emitting elements are alternately disposed between two adjacent first light-emitting elements. In this way, the brightness difference between the periphery and the center of the display region of the display device may be reduced, so as to alleviate the problem of non-uniform brightness of the display device. In addition, according to some embodiments, the display device may further include a light conversion layer disposed on the lower surface of the optical layer. Disposing the light conversion layer on the edge of the lower surface of the optical layer may improve the light conversion efficiency of the second light-emitting elements located in the non-display region of the display device, so as to prevent the periphery of the display region of the display device from displaying unwanted colors.

(52) Although some embodiments of the present disclosure and their advantages have been disclosed above, it should be understood that a person of ordinary skill in the art may change, replace and/or modify the present disclosure without departing from the spirit and scope of the present disclosure. The features between the embodiments of the present disclosure may be arbitrarily combined as long as they do not violate or conflict with the spirit of the disclosure. In addition, the scope of the present disclosure is not limited thereto the process, machine, manufacturing, material composition, device, method, and step in the specific embodiments described in the specification. A person of ordinary skill in the art will understand current and future process, machine, manufacturing, material composition, device, method, and step from the content disclosed in the present disclosure, as long as the current or future process, machine, manufacturing, material composition, device, method, and step performs substantially the same functions or obtain substantially the same results as the present disclosure. Therefore, the scope of the present disclosure includes the above-mentioned process, machine, manufacturing, material composition, device, method, and steps. The scope of the present disclosure should be determined by the scope of the claims. It is not necessary for any embodiment or claim of the present disclosure to achieve all of the objects, advantages, and/or features disclosed herein.

Claims

1. A display device, comprising: a substrate; a light-shielding layer disposed on the substrate; a plurality of first light-emitting elements disposed on the substrate; a plurality of second light-emitting elements disposed between the light-shielding layer and the substrate, wherein one of the plurality of second light-emitting elements is alternately disposed between any two adjacent first

light-emitting elements; an optical layer disposed on the plurality of first light-emitting elements and the plurality of second light-emitting elements; a color conversion layer disposed on the plurality of first light-emitting elements and the plurality of second light-emitting elements; a light conversion layer that partially overlaps the optical layer, and exposing at least a portion of the plurality of first light-emitting elements; and a liquid-crystal layer, wherein the optical layer, the color conversion layer, and the light conversion layer are disposed between the substrate and the liquid-crystal layer.

2. The display device as claimed in claim 1, wherein the two adjacent first light-emitting elements have a first extension line in a first direction, the second light-emitting element has a second extension line in a second direction, and the first extension line intersects the second extension line.

3. The display device as claimed in claim 2, wherein the first extension line and the second extension line are perpendicular to each other.

4. The display device as claimed in claim 1, wherein the plurality of second light-emitting elements are disposed to surround the plurality of first light-emitting elements.

5. The display device as claimed in claim 1, wherein the plurality of second light-emitting elements are disposed on at least one side of the plurality of first light-emitting elements.

6. The display device as claimed in claim 1, wherein the light conversion layer is disposed between the plurality of second light-emitting elements and the optical layer.

7. The display device as claimed in claim 1, wherein the light conversion layer is disposed between some of the plurality of first light-emitting elements and the optical layer.

8. The display device as claimed in claim 1, wherein the optical layer has a single-layer structure or a multi-layer structure.

9. The display device as claimed in claim 1, wherein the light conversion layer is further disposed on an upper surface of the optical layer.

10. The display device as claimed in claim 9, wherein the light conversion layer disposed on the upper surface of the optical layer is aligned with the light conversion layer that partially overlaps the optical layer in a normal direction of the substrate.

11. The display device as claimed in claim 1, wherein an outer edge of the light conversion layer is aligned with an edge of the optical layer in a normal direction of the substrate.

12. The display device as claimed in claim 1, further comprising a frame member surrounding the plurality of first light-emitting elements and the plurality of second light-emitting elements, wherein a portion of the optical layer and a portion of the light conversion layer are disposed on the frame member and are in contact with the frame member.

13. The display device as claimed in claim 1, wherein the substrate comprises a display region and a non-display region, the light-shielding layer covers the non-display region, the plurality of first light-emitting elements are disposed in the display region, and the plurality of second light-emitting elements are disposed in the non-display region.

14. The display device as claimed in claim 13, wherein sidewalls of the plurality of second light-emitting elements are aligned with a boundary of the non-display region.

15. The display device as claimed in claim 1, further comprising an adhesive layer formed on the substrate, the plurality of first light-emitting elements, and the plurality of second light-emitting elements, wherein the adhesive layer encapsulates the plurality of first light-emitting elements and the plurality of second light-emitting elements.

16. The display device as claimed in claim 1, wherein the plurality of first light-emitting elements located in the same column or row and the plurality of first light-emitting elements adjacent to one column or adjacent to one row are staggered to each other.

17. The display device as claimed in claim 1, wherein a first pitch between two adjacent first light-emitting elements and a second pitch between two adjacent second light-emitting elements are the same.

18. The display device as claimed in claim 17, wherein a third pitch between a first light-emitting

element and an adjacent second light-emitting element is $\frac{1}{2}$ of the first pitch and the second pitch.

19. The display device as claimed in claim 1, further comprising a brightness enhancement film disposed on the plurality of first light-emitting elements and the plurality of second light-emitting elements.
